# Lecture Scribe Bayes Theorem

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L5_A_S2.pdf
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You are required to generate a LECTURE SCRIBE intended strictly as EXAM-ORIENTED REFERENCE MATERIAL.

Use ONLY the provided context (lecture slides and permitted textbook excerpts).
Do NOT introduce any new explanations, intuition, examples, assumptions, or interpretations beyond what explicitly appears in the context.

The lecture scribe must be faithful, formal, and complete enough that a student can revise the lecture using only this document.

The scribe MUST include, wherever they appear in the context:
• Definitions and notation
• Assumptions and conditions
• Statements of results, propositions, or theorems
• Proofs or derivations written step by step
• Worked examples with all intermediate steps shown

The scribe MUST NOT:
• Add intuition, commentary, or simplifications
• Introduce new examples or alternative methods
• Reorder concepts beyond the lecture's logical flow
• Summarize or paraphrase loosely

Follow the structure demonstrated in the example lecture scribe below.
Match its organization, tone, and level of detail exactly.
Do not copy its content—use it only as a structural reference.

EXAMPLE LECTURE SCRIBE (STRUCTURE ONLY)

Title of Lecture

1. Topic Name
   Definition:
   Formal definition exactly as stated.

   Explanation:
   Description following the lecture's presentation order.

2. Result / Proposition
   Statement:
   Formal statement.

   Proof / Derivation:
   Step 1:
   Step 2:
   ...
   Conclusion.

3. Example
   Problem statement as given.
   Solution with intermediate steps shown explicitly.

Using the SAME structure, generate a detailed lecture scribe for the provided context.
Ensure correctness, completeness, and strict adherence to the source material.

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# CSE400 – Fundamentals of Probability in Computing

## Lecture 5: Bayes' Theorem, Random Variables, and Probability Mass Function

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**Date:** January 20, 2026

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# 1. Bayes' Theorem

## 1.1 Weighted Average of Conditional Probabilities

**Definition and Setup:**

Let $A$ and $B$ be events. We may express $A$ as

$$A = AB \cup AB^c$$

for, in order for an outcome to be in $A$, it must either be in both $A$ and $B$, or be in $A$ but not in $B$.

**Derivation:**

Since $AB$ and $AB^c$ are mutually exclusive, by Axiom 3,

$$\Pr(A) = \Pr(AB) + \Pr(AB^c)$$

Using the definition of conditional probability,

$$\Pr(A) = \Pr(A \mid B)\Pr(B) + \Pr(A \mid B^c)\Pr(B^c)$$

$$= \Pr(A \mid B)\Pr(B) + \Pr(A \mid B^c)[1 - \Pr(B)]$$

**Result:**

The probability of event $A$ is a weighted average of the conditional probabilities, with weights given by the probability of the event on which it is conditioned.

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## 1.2 Learning by Example

**Example 3.1 (Part 1/2)**

**Problem Statement:**

An insurance company believes that people can be divided into two classes: those who are accident prone and those who are not.

- Probability an accident-prone person has an accident in one year: $0.4$

- Probability a non–accident-prone person has an accident in one year: $0.2$
- Proportion of accident-prone people in the population: $0.3$

What is the probability that a new policyholder will have an accident within a year of purchasing a policy?

**Solution:**
Let

- $A_1$: event that the policyholder has an accident within a year
- $A$: event that the policyholder is accident prone

Then,

$$\begin{aligned}\Pr(A_1) &= \Pr(A_1 \mid A)\Pr(A) + \Pr(A_1 \mid A^c)\Pr(A^c) \\ &= (0.4)(0.3) + (0.2)(0.7) \\ &= 0.12 + 0.14 = 0.26\end{aligned}$$

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**Example 3.1 (Part 2/2)**
**Problem Statement:**
Suppose that a new policyholder has an accident within a year of purchasing a policy. What is the probability that he or she is accident prone?

**Solution:**
The desired probability is

$$\begin{aligned}\Pr(A \mid A_1) &= \frac{\Pr(A \cap A_1)}{\Pr(A_1)} \\ &= \frac{\Pr(A)\Pr(A_1 \mid A)}{\Pr(A_1)} \\ &= \frac{(0.3)(0.4)}{0.26} \\ &= \frac{6}{13}\end{aligned}$$

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## 1.3 Formal Introduction: Law of Total Probability and Bayes' Theorem

**Law of Total Probability (Formula 3.4)**
Let $B_1, B_2, \ldots, B_n$ be mutually exclusive and exhaustive events. Then,

$$\Pr(A) = \sum_{i=1}^{n} \Pr(AB_i)$$

Using

$$\Pr(AB_i) = \Pr(B_i \mid A) \Pr(A)$$

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**Bayes' Formula (Proposition 3.1)**
From the Law of Total Probability,

$$\Pr(B_i \mid A) = \frac{\Pr(A \mid B_i) \Pr(B_i)}{\sum_{j=1}^{n} \Pr(A \mid B_j) \Pr(B_j)}$$

**Terminology:**

- $\Pr(B_i)$: *a priori probability*
- $\Pr(B_i \mid A)$: *posterior probability*

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## 1.4 Learning by Example

**Example 3.2**
**Problem Statement:**
There are three cards:

- One card is red on both sides (RR)
- One card is black on both sides (BB)
- One card is red on one side and black on the other (RB)

A card is randomly selected and placed on the ground. If the upper side is red, what is the probability that the other side is black?

**Solution:**
Let

- $RR, BB, RB$: events corresponding to the selected card
- $R$: event that the upturned side is red

Then,

$$\Pr(RB \mid R) = \frac{\Pr(R \mid RB)\Pr(RB)}{\Pr(R \mid RR)\Pr(RR) + \Pr(R \mid RB)\Pr(RB) + \Pr(R \mid BB)\Pr(BB)}$$

Substituting values,

$$= \frac{(1/2)(1/3)}{(1)(1/3) + (1/2)(1/3) + (0)(1/3)}$$

$$= \frac{1/6}{1/2} = \frac{1}{3}$$

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## 2. Random Variables

### 2.1 Motivation and Concept

When an experiment is performed, we are frequently interested mainly in some function of the outcome rather than the outcome itself.

Examples described in lecture:

- Dice tossing: interest in the sum of two dice
- Coin flipping: interest in the total number of heads

**Definition:**
A *random variable* is a real-valued function defined on the sample space.

- Values are determined by outcomes of an experiment
- Probabilities are assigned to possible values of random variables

The distribution of a random variable can be visualized as a bar diagram where:

- x-axis represents values of the random variable
- Height of the bar at value $a$ is $\Pr[X = a]$

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### 2.2 Examples

**Example 1: Tossing 3 Fair Coins**

**Problem Statement:**

Suppose an experiment consists of tossing 3 fair coins. Let $Y$ denote the number of heads that appear.

**Possible Values:**

$$Y \in \{0, 1, 2, 3\}$$

**Probabilities:**

$$\Pr(Y = 0) = \Pr\{(t, t, t)\} = \frac{1}{8}$$

$$\Pr(Y = 1) = \Pr\{(t, t, h), (t, h, t), (h, t, t)\} = \frac{3}{8}$$

$$\Pr(Y = 2) = \Pr\{(t, h, h), (h, t, h), (h, h, t)\} = \frac{3}{8}$$

$$\Pr(Y = 3) = \Pr\{(h, h, h)\} = \frac{1}{8}$$

Since $Y$ must take one of these values,

$$\sum_{y=0}^{3} \Pr(Y = y) = 1$$

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## 3. Probability Mass Function (PMF)

### 3.1 Concept

A random variable that can take on at most a countable number of possible values is said to be *discrete*.

Let $X$ be a discrete random variable with range

$$R_X = \{x_1, x_2, x_3, \dots\}$$

(finite or countably infinite).

**Definition:**

The function

$$p_X(x_k) = \Pr(X = x_k)$$

is called the *Probability Mass Function (PMF)* of $X$.

Since $X$ must take on one of the values $x_k$,

$$\sum_k p_X(x_k) = 1$$

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## 3.2 Example: Two Independent Tosses of a Fair Coin

(As presented in lecture slides.)

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## 3.3 Example

**Problem Statement:**

The probability mass function of a random variable $X$ is given by

$$p(i) = c\lambda^i, \quad i = 0, 1, 2, \ldots$$

where $\lambda > 0$. Find:

- $\Pr(X = 0)$
- $\Pr(X > 2)$

**Solution:**

Since $p(i)$ is a PMF,

$$\sum_{i=0}^{\infty} c\lambda^i = 1$$

Using the geometric series,

$$c\sum_{i=0}^{\infty} \lambda^i = c \cdot \frac{1}{1-\lambda} = 1$$

$$\Rightarrow c = 1 - \lambda$$

Then,

$$\Pr(X = 0) = p(0) = c = 1 - \lambda$$

Also,

$$\Pr(X > 2) = 1 - \Pr(X \leq 2)$$
$$= 1 - [p(0) + p(1) + p(2)]$$

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**End of Lecture 5 Scribe**